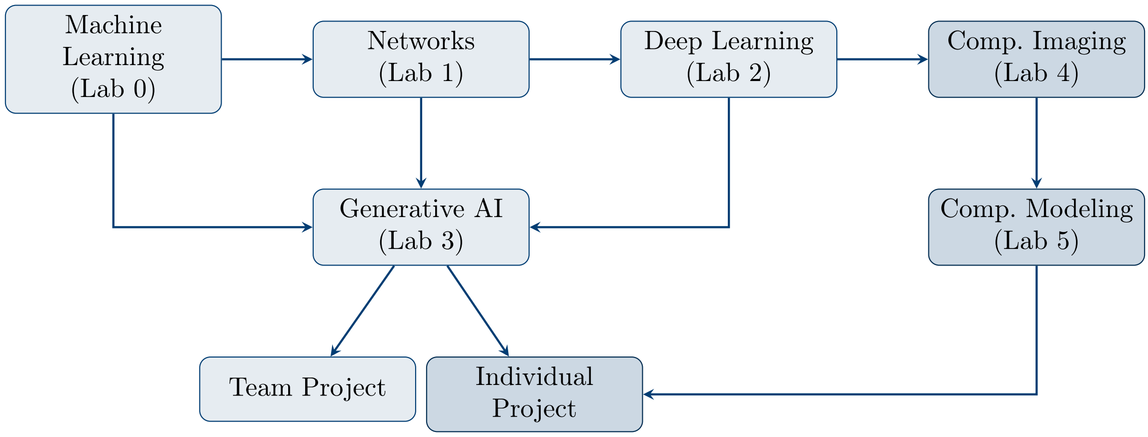
BMED365

Computational Imaging, Modeling and AI in Biomedicine

Course Description and Topic List

Spring 2026



Department of Biomedicine
Faculty of Medicine
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In collaboration with
Mohn Medical Imaging and Visualization Centre (MMIV)

Course Responsible: Arvid Lundervold

Course Material: https://github.com/arvidl/BMED365-2026

December 2025

Contents

1 Introduction and Motivation

1.1 Why AI in Medicine?

Artificial intelligence (AI) and machine learning are increasingly transforming healthcare and biomedical research (Topol, 2019). From image diagnostics and disease progression prediction to clinical decision support and drug development – AI-based tools are becoming increasingly relevant for healthcare professionals and researchers.

Core Questions of the Course

- How do machine learning and deep learning work?
- How can AI be applied responsibly in medicine?
- What are the opportunities and limitations of current AI technology?
- What does *precision medicine* mean in practice?

1.2 The Course's Place in the Curriculum

BMED365 is an elective course worth 10 ECTS credits offered to students with interest in AI in a health context. The course consists of two blocks: Block 1 (January) is shared with ELMED219, and Block 2 (February) includes additional labs on computational imaging and modeling.

1.3 Learning Outcomes

Upon completion of the course, the student should be able to:

Knowledge

- Explain fundamental principles of machine learning, deep learning, and generative AI
- Describe network analysis and patient similarity networks (PSN)
- Account for ethical and regulatory aspects of medical AI
- Understand concepts such as *explainable AI* (XAI) and *trustworthy AI*

Skills

- Apply Python and Jupyter Notebooks for simple ML analyses
- Use AI tools (ChatGPT, Gemini, Claude) as learning and work partners
- Construct and analyze patient similarity networks
- Write a research plan using LaTeX

General Competence

- Assess opportunities and limitations of AI in clinical practice
- Reflect on ethical implications of AI technology
- Collaborate interdisciplinarily in project groups
- Communicate professionally about AI-related topics

2 Course Structure and Content

2.1 Overview of Activities

The course is organized around four laboratories (Labs), a quick start course, and a team project. Table 1 provides an overview.

Table 1: Overview of course activities

Activity	Topic	Description	Hours
Quick Start	AI-assisted Python	Intro to Python and AI tools	4
Lab 0	Machine Learning	ML with scikit-learn and PyCaret	6
Lab 1	Network Science	Graph theory and PSN with NetworkX	6
Lab 2	Deep Learning	CNN and neural networks with PyTorch	8
Lab 3	Generative AI & LLM	Transformers, prompting, ethics	6
Team Project	Precision Medicine	Research plan for glioblastoma	15

2.2 Quick Start: AI-Assisted Python Programming

The quick start course is designed for students without prior programming experience. It provides a practical introduction to:

- Google Colab and Jupyter Notebooks
- Python basics: variables, data types, lists, functions
- AI as a programming partner (Gemini/ChatGPT)
- Preview of Lab 0 (classification) and Lab 1 (networks)

2.3 Lab 0: Machine Learning (ML)

Lab 0 provides a practical introduction to machine learning with focus on classification and evaluation.

Table 2: Notebooks in Lab 0

No.	Notebook	Content
01	Simple examples	Features, labels, training, testing, decision trees
02	Binary classification	Confusion matrix, ROC, precision, recall, F1-score
03	PyCaret quick guide	AutoML for rapid prototyping

Learning Objectives:

- Distinguish between supervised and unsupervised learning
- Use train/test split and cross-validation
- Interpret evaluation metrics (accuracy, precision, recall, AUC)
- Understand TRIPOD guidelines for medical ML

2.4 Lab 1: Network Science and PSN

Lab 1 introduces graph theory and the concept of *patient similarity networks* (PSN) – an approach for identifying patient subgroups based on similarity.

Table 3: Notebooks in Lab 1

No.	Notebook	Content
00	Introduction	Graph theory, adjacency matrices, centrality measures
01	NetworkX tutorial	Create, manipulate, visualize graphs
02	PSN with IRIS	Similarity calculation, community detection
03	PSN with IBS data	Clinical application: brain and cognition
04	PSN with IQ data	WAIS-IV and cognitive profiles

Learning Objectives:

- Define graph, node, edge, and understand adjacency matrices
- Calculate centrality measures (degree, betweenness, eigenvector)
- Construct PSN from clinical data
- Apply the Louvain algorithm for community detection

2.5 Lab 2: Deep Learning (DL)

Lab 2 explores deep learning with focus on neural networks, CNN, and medical image analysis.

Table 4: Notebooks in Lab 2 (prioritized)

Part	Pri	Notebook	Content
A	1	CNN-intro	Conceptual intro to CNN
A	1	MNIST-MLP	Your first DL model
A	1	MNIST-CNN	CNN on handwritten digits
B	1	NN-intro	Biological vs. artificial neurons
B	1	Learning in NN	Backpropagation, gradient descent
B	1	Heart disease	Classification from tabular data
B	1	ECG-arrhythmia	CNN on ECG signals
C	2	CNN-architecture	Environment setup and architecture
C	2	CNN-training	Training and saving model
C	2	CNN-testing	Evaluation and Grad-CAM (XAI)
D	2	MRI-dementia	MRI image analysis for dementia

Priority levels: 1 = core, 2 = recommended, 3 = optional, 4 = advanced

Learning Objectives:

- Explain the structure of a neural network
- Understand backpropagation and gradient descent

- Build and train CNN with PyTorch
- Use Grad-CAM to interpret model decisions

2.6 Lab 3: Generative AI and Large Language Models

Lab 3 covers generative AI and large language models (LLM) with focus on medical applications, ethics, and explainability.

Table 5: Notebooks in Lab 3

No.	Priority	Topic	Time
01	Core	Introduction to generative AI	45 min
02	Core	Transformer architecture	60 min
03	Core	LLM fundamentals (tokens, temp.)	45 min
04	Core	Prompt engineering	90 min
05	Important	Explainable AI (XAI)	60 min
06	Important	AI ethics in medicine	60 min
07	Optional	Trustworthy AI	60 min
08	Optional	Neurosymbolic AI + Agentic AI	90 min
09	Optional	ChatGPT/Claude API	60 min

Note: Notebook 08 (Neurosymbolic AI) contains a comprehensive **case study on brain tumors (glioma)** that demonstrates:

- Neurosymbolic classification based on WHO 2021 CNS classification
- Integration of MRI analysis (neural) with knowledge graphs (symbolic)
- Agentic AI orchestrating clinical workflow

Learning Objectives:

- Explain self-attention and transformer architecture
- Understand the difference between *prompt engineering* and *context engineering*
- Apply prompt engineering techniques (zero-shot, few-shot, CoT)
- Apply model-specific prompting techniques for GPT-5.2, Claude Opus 4.5, and Gemini 3
- Evaluate XAI methods such as SHAP and LIME
- Discuss AI ethics, bias, and EU AI Act
- Understand neurosymbolic integration of neural and symbolic methods
- Describe agentic AI and its role in clinical decision support

2.7 Team Project: Precision Medicine for Glioblastoma

The team project is a central part of the course where students in interdisciplinary groups develop a research plan (grant proposal sketch) for a hypothetical project on AI and imaging for brain tumors.

Important about the team project

The task is to **write a research plan** – not to actually implement the project with coding or data analysis.

Focus Areas:

1. Imaging technologies and modalities (MRI, PET)

2. Image-derived biomarkers for glioblastoma
3. Machine learning techniques (segmentation, classification)
4. Data management plan and ethical considerations

Relevant Background: The Glioma Case Study

Notebook 08 in Lab 3 contains a comprehensive neurosymbolic AI case study for glioma classification based on WHO 2021 CNS classification. This demonstrates:

- Integration of MRI segmentation (BraTS) with molecular markers (IDH, MGMT, 1p/19q)
- Knowledge graphs for treatment recommendations
- Agentic AI for orchestrating clinical workflow

Students can use this as inspiration for the team project.

Report Structure:

- Research plan: 3–5 pages (background, objectives, methods, evaluation)
- Data management plan and ethics: 1.5–2.5 pages
- Written in English
- LaTeX template available on Overleaf

3 Detailed Topic List

This section provides a comprehensive topic list organized by theme. Topics are marked with priority: **C** = core (mandatory), **I** = important (recommended), **S** = supplementary (optional).

3.1 Machine Learning – Fundamental Concepts

Pri	No.	Topic
C	M01	Define *machine learning* and distinguish from traditional programming
C	M02	Explain the difference between *supervised* and *unsupervised* learning
C	M03	Describe the concepts of *features* (input) and *labels* (output)
C	M04	Explain why we split data into *training* and *test sets*
C	M05	Define *overfitting* and *underfitting*
C	M06	Understand *bias-variance trade-off*
I	M07	Describe k-fold *cross-validation* and its advantages
I	M08	Explain what a *baseline model* is and why it is important
C	M09	Distinguish between *classification* and *regression*
I	M10	Know simple models: decision tree, k-NN, logistic regression

3.2 Evaluation of ML Models

Pri	No.	Topic
C	E01	Interpret a *confusion matrix*
C	E02	Define TP, TN, FP, FN in medical context
C	E03	Calculate and interpret *accuracy*
C	E04	Calculate and interpret *precision* (PPV)
C	E05	Calculate and interpret *recall/sensitivity*
C	E06	Calculate and interpret *specificity*
I	E07	Calculate and interpret *F1-score*
C	E08	Understand ROC curve and AUC as measures of model quality
I	E09	Explain when accuracy is insufficient (imbalanced datasets)
I	E10	Know TRIPOD guidelines for reporting

3.3 Graph Theory and Network Science

Pri	No.	Topic
C	N01	Define what a *graph* is (nodes and edges)
C	N02	Distinguish between *directed* and *undirected* graphs
C	N03	Distinguish between *weighted* and *unweighted* graphs
C	N04	Represent a graph as an *adjacency matrix*
C	N05	Calculate *degree centrality* and interpret the result
I	N06	Calculate *betweenness centrality* and interpret the result
I	N07	Calculate *eigenvector centrality* and interpret the result
I	N08	Explain *clustering coefficient*
I	N09	Describe *community detection* and the Louvain algorithm
C	N10	Know the NetworkX library for Python

3.4 Patient Similarity Networks (PSN)

Pri	No.	Topic
C	P01	Explain the concept of *patient similarity networks* (PSN)
C	P02	Describe how PSN can support *precision medicine*
C	P03	Calculate similarity between patients (Euclidean, Manhattan, Gower)
I	P04	Construct a PSN from a patient-feature matrix
I	P05	Choose threshold value for edge creation in PSN
I	P06	Identify patient subgroups via community detection
S	P07	Discuss advantages and limitations of the PSN approach
S	P08	Know about multimodal PSN (integration of different data types)

3.5 Neural Networks and Deep Learning

Pri	**No.**	**Topic**
C	D01	Compare biological and artificial neurons
C	D02	Describe the structure of a *multilayer perceptron* (MLP)
C	D03	Explain what an *activation function* is (ReLU, sigmoid)
C	D04	Understand the concept of *forward propagation*
C	D05	Explain *backpropagation* at a conceptual level
C	D06	Understand *gradient descent* and learning rate
I	D07	Know about *loss functions* (cross-entropy, MSE)
C	D08	Describe a *convolutional neural network* (CNN)
C	D09	Explain what a *convolution filter* does
C	D10	Describe *pooling layers* and their function
I	D11	Know about *batch normalization* and *dropout*
I	D12	Understand the concept of *transfer learning*
S	D13	Know about advanced architectures (ResNet, ViT)

3.6 Generative AI and Large Language Models

Pri	**No.**	**Topic**
C	G01	Define *generative AI* and distinguish from discriminative AI
C	G02	Explain the *self-attention* mechanism at a conceptual level
C	G03	Describe the transformer architecture (encoder-decoder)
C	G04	Explain what *tokens* are and how tokenization works
C	G05	Understand the concept of *context window*
C	G06	Explain what *temperature* means in text generation
C	G07	Apply *zero-shot* prompting
C	G08	Apply *few-shot* prompting
I	G09	Apply *Chain-of-Thought* (CoT) prompting
I	G10	Describe best practices for *system prompts*
C	G11	Define *hallucination* and its implications
I	G12	Know about GPT-5.2, Claude Opus 4.5, Gemini 3 and their strengths
S	G13	Explain the concept of *foundation model*
S	G14	Describe RAG (Retrieval-Augmented Generation)
I	G15	Understand the difference between *prompt engineering* and *context engineering*
I	G16	Describe components of context engineering (system prompt, RAG, tools, conversation history)
I	G17	Apply model-specific prompting techniques for GPT-5.2, Claude, and Gemini

I	G18	Handle uncertainty and hallucination risk in medical prompts
S	G19	Use structured extraction from medical documents (JSON schema)
S	G20	Compare model strengths for different medical tasks

3.7 Explainable AI (XAI)

Pri	**No.**	**Topic**
C	X01	Explain why explainability is important in medical AI
C	X02	Distinguish between *global* and *local* explainability
I	X03	Distinguish between *ante-hoc* and *post-hoc* explainability
I	X04	Describe SHAP (SHapley Additive exPlanations)
I	X05	Describe LIME (Local Interpretable Model-agnostic Explanations)
I	X06	Explain Grad-CAM for CNN visualization
I	X07	Discuss limitations of XAI methods
S	X08	Know about attention visualization in LLM

3.8 Trustworthy AI and Robustness

Pri	**No.**	**Topic**
C	T01	Define *trustworthy AI* according to EU guidelines
I	T02	Explain the concept of *robustness* in ML
I	T03	Describe *distributional shift* and its consequences
I	T04	Explain the difference between epistemic and aleatoric uncertainty
I	T05	Describe *human-in-the-loop* (HITL) systems
I	T06	Discuss the importance of continuous monitoring
S	T07	Know about adversarial attacks

3.9 AI Ethics and Regulation

Pri	**No.**	**Topic**
C	A01	Discuss the four bioethical principles in AI context
C	A02	Identify types of bias in medical AI systems
C	A03	Explain privacy concerns when using LLM
C	A04	Describe main features of EU AI Act
I	A05	Explain risk classification in EU AI Act
I	A06	Discuss requirements for high-risk AI in healthcare
C	A07	Reflect on responsibility distribution when AI fails
I	A08	Know about GDPR-relevant aspects of AI

S	A09	Discuss algorithmic fairness

3.10 Neurosymbolic AI and Agentic AI

Pri	No.	Topic
S	S01	Contrast symbolic and connectionist AI
S	S02	Explain the concept of *neurosymbolic integration*
S	S03	Describe what a *knowledge graph* is
S	S04	Know about medical ontologies (SNOMED CT, ICD, WHO CNS classification)
S	S05	Discuss potential advantages of neurosymbolic AI in medicine
S	S06	Describe how neurosymbolic AI can combine MRI analysis (neural) with WHO classification (symbolic) for glioma diagnostics
S	S07	Explain how knowledge graphs can validate neural predictions
S	S08	Define *agentic AI* and its core properties (planning, tool use, reflection)
S	S09	Describe how an AI agent can orchestrate a clinical workflow (PACS, literature search, biobank)
S	S10	Explain the concept of *human-in-the-loop* in agentic systems
S	S11	Discuss ethical challenges with autonomous AI agents in healthcare

3.11 Medical Image Analysis

Pri	No.	Topic
C	B01	Explain basic MRI principles at a high level
I	B02	Know about different MR sequences (T1, T2, FLAIR)
C	B03	Describe *segmentation* in medical image analysis
I	B04	Know about the BraTS challenge for brain tumor segmentation
I	B05	Describe radiomic features and quantitative imaging
S	B06	Know about nnU-Net and MONAI as tools

3.12 Practical Skills

Pri	No.	Topic
C	F01	Run Jupyter Notebooks in Google Colab
C	F02	Use Python variables, lists, and simple functions
I	F03	Import and use libraries (numpy, pandas, matplotlib)

I	F04	Read and inspect datasets with pandas
I	F05	Train a simple model with scikit-learn
I	F06	Visualize results with matplotlib
I	F07	Use NetworkX for simple network analysis
S	F08	Build and train a model with PyTorch
C	F09	Use AI tools (ChatGPT, Claude, Gemini) as coding help
I	F10	Write simple documents with LaTeX/Overleaf
I	F11	Compose effective prompts for medical tasks
I	F12	Apply context engineering with system prompts, RAG, and tools
I	F13	Choose the right LLM model based on task type
S	F14	Compare output from different LLMs for quality assessment

4 Tools and Resources

4.1 Software and Platforms

Table 18: Software and tools used in the course

Tool	Type	Use Area
Google Colab	Platform	Run Jupyter Notebooks in browser
Python 3.10+	Programming language	All coding in the course
scikit-learn	Library	Machine learning (Lab 0)
NetworkX	Library	Network analysis (Lab 1)
PyTorch	Framework	Deep learning (Lab 2)
tiktoken	Library	Tokenization (Lab 3)
PyCaret	Library	AutoML (Lab 0)
Overleaf	Platform	LaTeX editing (Team project)

4.2 AI Assistants

Table 19: AI tools available for students

Tool	Access	Use Area
ChatGPT	Free/Plus	General AI assistant, coding help
Claude	Free/Pro	Text analysis, academic writing
Gemini	Built into Colab	Code generation and explanation
NotebookLM	Free (Google)	Document analysis
chat.uib.no	UiB login	UiB internal AI service
Medical AI GPT	ChatGPT Plus	Customized for BMED365

4.3 Recommended Reading

Books and resources (freely available):

- Barabási (2016): *Network Science* – http://networksciencebook.com
- Molnar (2022): *Interpretable Machine Learning* – https://christophm.github.io/interpretable-ml-b
- Goodfellow et al. (2016): *Deep Learning* – https://www.deeplearningbook.org

Key articles:

- Lundervold and Lundervold (2019): Deep learning in medical image processing
- Vaswani et al. (2017): Transformer architecture
- Singhal et al. (2023): LLM and clinical knowledge
- Morley et al. (2020): Ethics in AI and health
- Garcez and Lamb (2020): Neurosymbolic AI – the third wave
- Louis et al. (2021): WHO 2021 CNS classification (gliomas)

Advanced (optional):

- Nicholson and Greene (2020): Knowledge graphs in biomedicine
- Xi et al. (2023): Agentic AI with large language models

5 Assessment and Exam

5.1 Assessment Forms

The course is assessed with pass/fail based on:

1. **Team project (group-based):** Research plan as described in section 2.7

2. **Digital home exam (individual):** 2-hour exam on Inspera

5.2 Exam Format

- 8 multiple choice questions (MCQ) with 5 alternatives, 2 correct answers per question
- 2 free-text essay questions
- All aids allowed (must be specified in the answer)
- AI tools allowed to understand and explain medical AI

5.3 Exam Topics

The following topics may be covered in the exam:

- Multimodal data
- Basic machine learning
- Open science
- MRI technology
- Generative AI
- Large language models (LLM)
- Biomarkers
- Deep learning and overfitting
- Patient similarity networks (PSN)
- The equation $y \approx f(\mathbf{X}, \theta)$
- ML in diagnostics
- AI ethics and regulation
- Neurosymbolic AI
- Agentic AI

6 Preliminary Schedule

Table 20: Schedule BMED365 – January-February 2026

Week	Date	Time	Activity	Location
2	Mon 05.01	10:15–14:00	Intro, SW installation	Hist 1
2	Wed 07.01	14:15–16:00	Tools, team, Lab 0	Hist 1
2	Fri 09.01	10:15–13:00	Lab 0, Lab 1	Hist 1
3	Tue 13.01	09:15–13:00	Quick Start AI-Python	Hist 1
3	Fri 16.01	08:15–13:00	Lab 2 (Deep Learning)	Hist 1
4	Tue 20.01	08:15–12:00	Lab 3 (GenAI+LLM)	Hist 1
4	Tue 20.01	13:15–16:00	Team project guidance	Hist 1
5	Tue 27.01	08:15–10:00	Team presentations	Hist 1
5	Fri 30.01	11:00–13:00	**HOME EXAM**	Inspera

7 Contact Information

Course Responsible: Arvid Lundervold, Department of Biomedicine, UiB
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Course Material: https://github.com/arvidl/BMED365-2026

MittUiB: https://mitt.uib.no/courses/50716 (enrolled only)

References

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A Illustrations

A.1 Course Structure

A.2 AI in Medicine – Main Topics

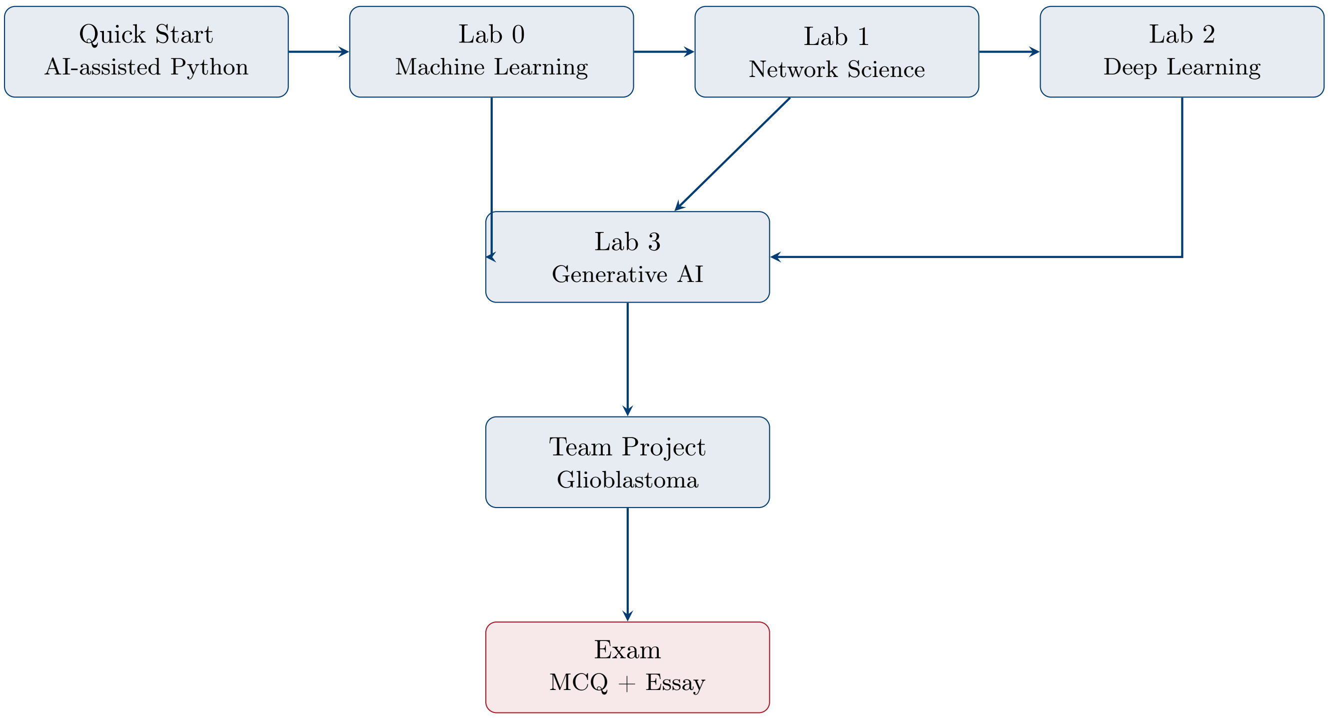

Figure 1: Course structure and progression in BMED365

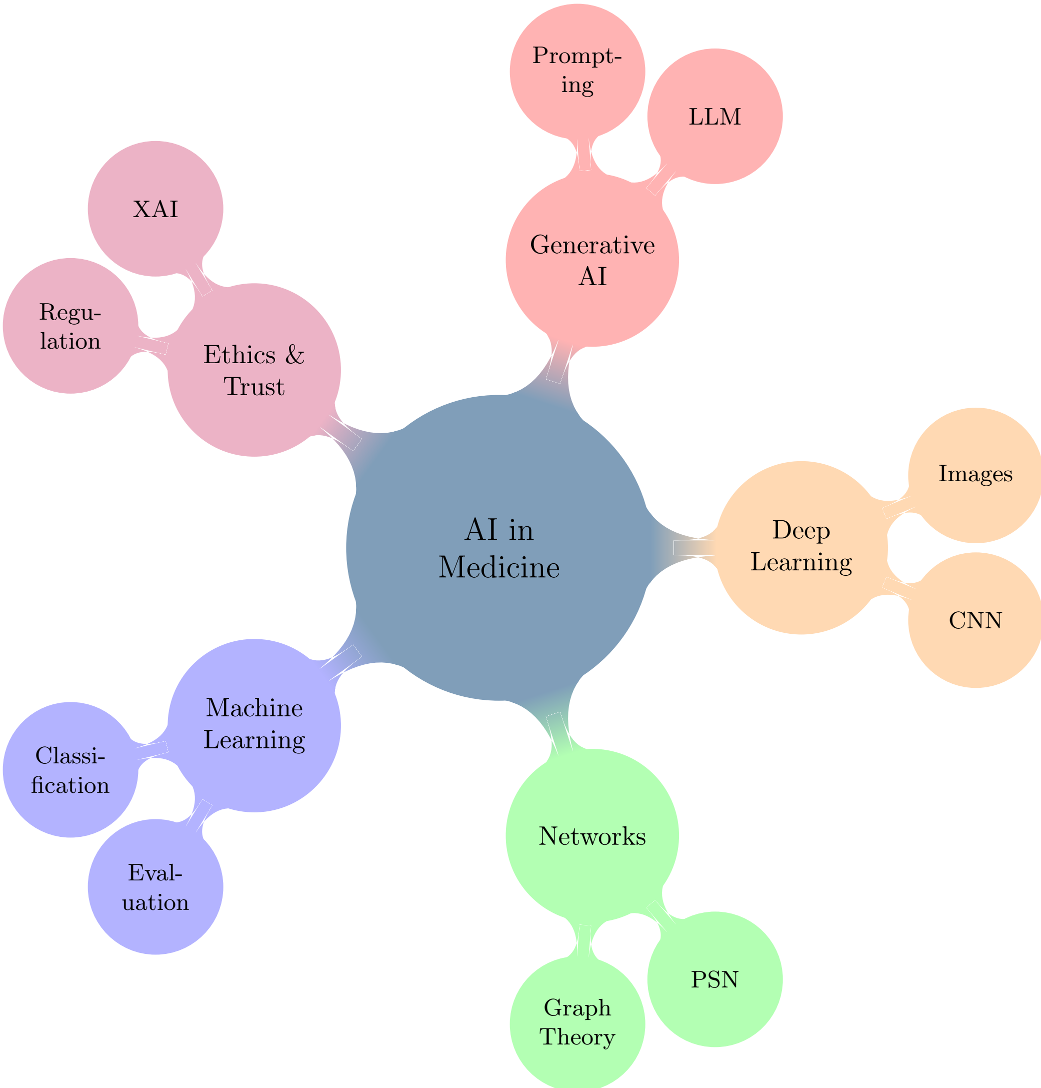

Figure 2: Main topics in BMED365: Computational Imaging, Modeling and AI in Biomedicine